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1 Abstract

Objective: To review the plausibility of insect detection of infrared (IR) vibrational signatures of semiochemicals through electromagnetic coupling in addition to molecular binding, and to produce falsifiable predictions through the integration of comparative entomology, spectroscopy, neural timing analysis, and computational electromagnetism.

Methods: We integrate: (i) morphometric analysis of antennal sensilla across 500+ specimens from diverse taxa, (ii) ATR-FTIR spectroscopy of cuticular hydrocarbons with sub-micron resolution ($\pm 0.1\ \mu$m), (iii) meta-analysis of olfactory receptor neuron response latencies, and (iv) deterministic electromagnetic models validated against antenna theory. All analyses use fixed random seeds (42) for reproducibility, and all code to generate results and this manuscript is available in the open source repo. We also provide a mathematical appendix with detailed derivations and computational implementations.

Results: Antenna sensilla dimensions (typically 2–160 μm across sensillum types; e.g., trichodea 6–160 μm, basiconica 2–8 μm, coeloconica 5–15 μm) show correlation with predicted IR resonances across diverse insect taxa. Modeled electromagnetic detection achieves 1–5 ms latencies (minimum ~3 ms reported), faster than diffusion-based molecular binding. CHC vibrational spectra overlap Earth's atmospheric transmission windows (2–5, 8–14, 17–25 μm), facilitating detection ranges of possibly tens of meters. Integrated case studies validate predictions and error bounds across all tested scenarios, supported by peer-reviewed evidence from multiple domains.

Conclusions: Our framework generates three key, falsifiable predictions related to the insect olfactory system: (1) wavelength-specific sensilla tuning where sensilla dimensions predict IR resonance frequencies (testable through imaging and electrophysiology), (2) IR-only behavioral orientation without volatile chemical cues (testable via controlled IR stimulation in olfactometers), and (3) sub-10 ms neural responses to narrowband IR stimulation distinguishable from thermal responses (testable via single-sensillum recordings with thermal controls). We provide protocols for distinguishing electromagnetic detection from thermal artifacts using matched power deposition and wavelength specificity.

Implications: Applications of this work include next-generation biomimetic sensors, precision pest management, and fundamental advances in sensory biology.

Keywords: insect olfaction, infrared detection, vibrational theory, electromagnetic sensing, sensilla morphology, cuticular hydrocarbons, atmospheric transmission, biomimetic sensors

Reproducibility: Complete implementation with seven case studies in Appendices (sec:app$_{s}ensilla_{a}rray, sec : app_{e}$